From force diagrams to motion

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use a force diagram to predict acceleration and velocity change.

SUCCESS CRITERIA - I can...

1. Write a force sum along each chosen axis.
2. Connect net force and mass to acceleration.
3. Use velocity direction to decide the initial speed change.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “parallel” and “perpendicular” to explain your answer. Sentence starters are optional.

parallel: along the same direction as a line or surface

perpendicular: at a right angle, 90°, to a line or surface

component: the part of a vector along one chosen axis

01 Warm-up: bring back an earlier idea

A **3 kg** object has **22 N** right and **7 N** left. **Draw the forces** and **find net force** and acceleration.

Word help: acceleration = rate of change of velocity

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

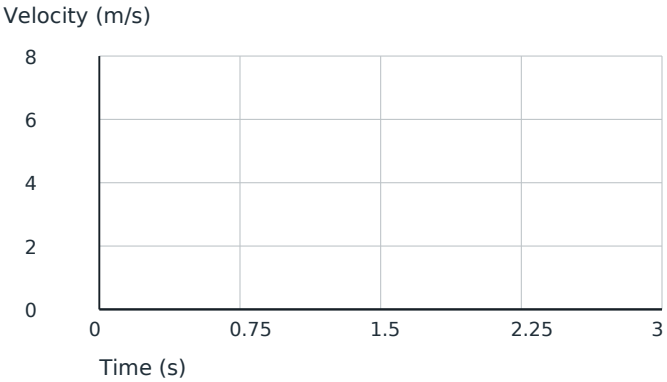
02 Worked example: explain the prediction

Reference: $a_x = \Sigma F_x/m$; $a_y = \Sigma F_y/m$.
Choose axes first. CPA ramp problems supply the along-ramp gravity component; no trigonometry is required.

Worked reasoning: A **2 kg** cart initially at rest has a constant **4 N** net force right for **3 s**. Acceleration is **2 m/s²**; final speed is **6 m/s** right.

Your turn: Draw a force diagram and a velocity-time graph that agree. **Mark where mass enters the** reasoning.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .



Marzano check - circle: ① ② ③

03 Horizontal case

The diagram shows forces on a **5 kg** box. **Find net horizontal force** and acceleration.

Sentence starter: I found ... [unit]. This value means ... in this situation.



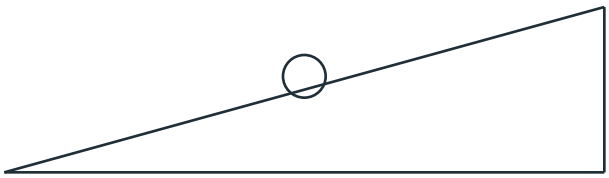
The dot represents the object. Mass: 5 kg

Marzano check - circle: ① ② ③

04 Use a given ramp component

A **2 kg** block slides down a ramp. Gravity’s down-ramp component is **10 N**; friction is **4 N** up the ramp. **Find net force** and acceleration along the ramp.

Sentence starter: I found ... [unit]. This value means ... in this situation.



10 N down; 4 N up. The marker represents the object on the sloping surface.

Marzano check - circle: ① ② ③

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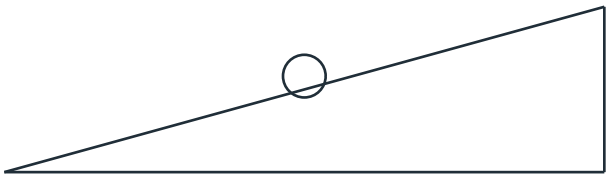
Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Moving up, accelerating down

A block slides up a rough ramp after the push ends. **Draw its velocity separately** from the along-ramp forces. Gravity's along-ramp component and friction both point down the ramp. **Explain what happens** to the block's speed.

Sentence starter: I think ... . I can point to ..., which shows ..., so ... .



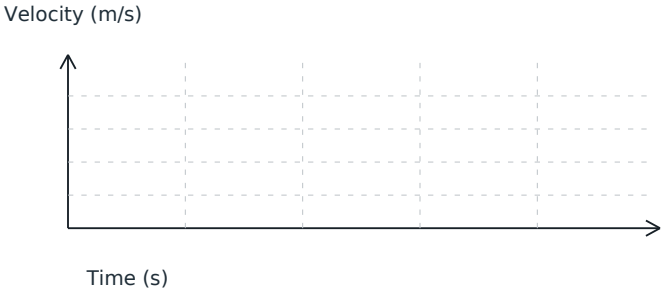
Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A velocity graph is constant at **-5 m/s** for **7 s**. **Find displacement**. A second object has components **9 m/s** *east* and **4 m/s** *south*; **sketch its direction**. **Sketch the velocity**–time graph; **label the given values**.

Word help: displacement = change from initial position to final position, with direction

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: 1 2 3

07 Exit check

A **3 kg** cart moves left. It has **15 N** right and **6 N** left.
1. **Find acceleration**.
2. Does its leftward speed initially increase or decrease?

Sentence starter: I found ... [unit]. This value means ... in this situation.



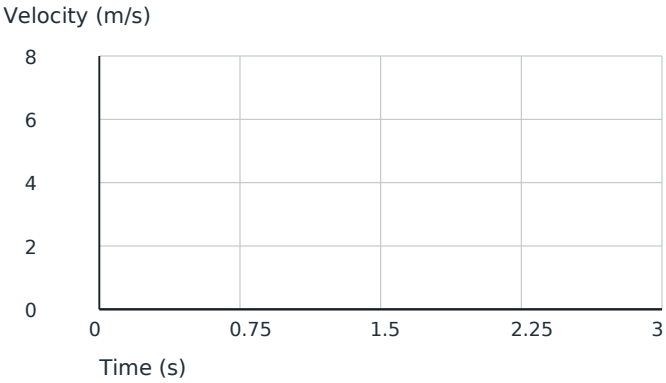
3 kg cart; current velocity is leftward.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Return to the worked example in problem **2**.
1. Keep the **4 N** net force and **3 s** time but double the mass.
2. **Compare the two velocity-time graphs** and final speeds.

Sentence starter: Both ... . They differ in ... because ... .



Marzano check - circle: 1 2 3